

CNC AI Quotation Generator: An Optimization-Driven Intelligent Framework

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Abstract

The CNC machining industry frequently uses a manual process to quote for machining, leading to inconsistencies, slow response times, and a reliance on the expertise of the operator's experience as it relates to cost estimation. Most of the tools currently available do not provide both CAD visualization and basic cost calculations, and as a result, they contain no are either no integrated optimization tools or are no intelligent manufacturability analysis tools. The CNC AI Quotation Generator uses DXF files to automatically process them into quotes with a high degree of accuracy for the machining of parts. The CNC AI Quotation Generator will provide the ability to optimize production via the use of geometric entity extraction, complexity scoring, material aware cost modeling, TSP based cutting path optimization, and grid plate nesting to enable companies within the CNC machining industry to increase operating efficiency and material utilization. An AI advisory module is provided which assists by offering materials selection recommendations and manufacturing insights. The experimental evaluation indicates that estimation time has been reduced, path efficiency has increased, and cost accuracy has been maintained for all quotes evaluated, thus providing a highly scalable and intelligent solution for organizations in the modern-day CNC manufacturing environment.

Keywords

CNC machining, Cost estimation, DXF processing, Tool path optimization, TSP, Nesting optimization.

1. Introduction

The significance of precise machining cost estimation in various manufacturing processes with intense competition cannot be overemphasized. It is an extremely relevant part

in the price quotation, production planning, and profit estimation. The existing cost estimation methodologies are primarily based upon interpretation of engineering drawings and empirical knowledge [1]. These are highly non-scales able, inconsistent and time consuming especially when the product geometries are more complex, and customers are demanding tighter quotation deadlines. Several research works have been carried out by researchers to provide cost estimation methods based upon CAD data [2].

Feature based estimation is one of the methods devised to include CAD derived information in manufacturing cost estimation. However, the feature-based methods require a set of rules for different product geometries [3]. Further, tool path planning for CNC machining problem can be formulated as a Travelling Salesman Problem (TSP) that helps to find an efficient machining path which, in turn, minimizes the non-production travelling time [4]. Another area where optimization is required is sheet cutting operation, where poor nesting may lead to a significant amount of material waste [5].

Although such developments are valuable, there are still limited integrations of automated geometric extraction, cost modeling, path optimization, material nesting and AI-enabled manufacturability recommendations into a single quotation tool [6]. In this paper, we present a CNC AI Quotation Generator that uses DXF files to automate machining cost estimates, optimize for optimal performance, and generate professional quotations with AI-enabled manufacturability insights.

2. Background

Computer-Aided Process Planning (CAPP) systems were developed as a bridge between computer-aided design (CAD) and manufacturing execution by integrating design

data into a systematic flow of machining operations and production parameters [7]. CAPP systems provide the necessary support for systematic estimation of machining time, resource optimization and sequencing. However, most existing CAPP-based solutions rely on predefined feature libraries and rule-based feature-to-operation mappings, which are inflexible for arbitrary or irregular geometries, for instance extracted from 2D CAD files formats such as DXF.

Feature-based cost estimation approaches have been proposed to cover the degree of automation needed by identifying feature geometries and assigning their associated cost drivers and machining operations [8]. However, such methods are still limited in terms of manual tuning of parameters and optimization. In addition, the traditional cost estimation models are usually limited to the estimation of machining time and material consumption without integration of operational optimization into the quotation process.

CNC tool path planning can be formulated as combinatorial optimization problem from an optimization perspective, and the classical modeling is based on Traveling Salesman Problem (TSP) to minimize travel distance among machining features [9]. Sheet material cutting and layout planning are also similar to the nesting and bin-packing problems to maximize material utilization and minimize wastage [10]. These optimization methods are commonly used in manufacturing research, but they tend to be limited to the modular implementation instead of the integration into quotation systems.

Achieve a high integrated, optimization-based, AI-assisted framework by integrating geometric analysis, cost modeling and operational optimization into a single automatic quotation system.

3. Literature Review

The cost estimation for the manufacturing has been studied extensively, and our initial work of classification and evaluation of cost estimation methods was presented [1], where parametric models, feature-based costing, and empirical rule-based system were proposed. Neural network-based models were subsequently developed to enhance the prediction by learning from historical machining data [2]. These methods provide more accurate predictions, but still require large training data sets, and are usually difficult to interpret in an actual industrial environment.

Feature-based cost estimation methods add the geometric features extracted from CAD models to the manufacturing cost prediction pipeline [3]. Some of these feature-based methods bring the visibility of the manufacturing cost at

early design stages. However, they depend on the previously defined feature libraries and may not generalize well to alternative design structures. Concurrent engineering models tried to combine the cost estimation with manufacturing planning, but many of them are semi-automated and require expert intervention [5].

The research in computer-aided process planning (CAPP) focuses on systematic transformations of CAD models to machining operations and process parameters [6]. Automatic feature recognition methods are proposed to facilitate integration between the design and manufacturing systems [7]. Along with the development of the smart manufacturing paradigms, cyber-physical systems have facilitated tighter integration of the digital model and the physical production process [8]. Nevertheless, many of these systems focus on process planning rather than integrated quotation generation.

Optimization methods have also been extensively studied in the domain of manufacturing. The traveling salesman problem (TSP) is a commonly used framework that has been used to optimize tool travel distance in the machining process [9]. Genetic algorithms and swarm intelligence and other metaheuristic methods have been used for optimization of CNC tool paths to reduce non-productive times [10], [11]. These optimization methods are often run independently of the cost estimation module.

Nesting and bin-packing algorithms have been used to optimize material usage to reduce raw material waste in the sheet cutting process [12]. Review of packing algorithms have shown the importance and computational complexity of layout planning for industrial applications [13].

Nesting optimization is hardly incorporated in an automated quotation system. Artificial intelligence methods are increasingly used in smart manufacturing systems for decision support and predictive analytics [14]. Intelligent manufacturing systems review have highlighted the role of intelligent algorithms for manufacturing cost modeling, process optimization and decision making [15]. The majority of the research studies consider the cost estimation, optimization and intelligent advisory as independent modules, not as integrated modules.

From the previous review, it can be concluded that although considerable efforts have been made in cost modeling, process planning, tool path optimization, nesting approaches, and AI based manufacturing systems, there is hardly a single CNC quotation system with integrated automated geometric extraction, cost estimation, operational optimization, and intelligent advisory systems. This gap motivates the development of the proposed CNC AI Quotation Generator.

4. System Architecture

The present invention relates to a modular and layered system architecture that integrates CAD processing, cost modeling, optimization and AI-based advisory functionality into a single quotation system. The overall architecture is characterized by a data-processing pipeline where geometric data derived from uploaded DXF files is progressively transformed into cost data, optimization data and a quotation document. The system has been implemented as a web application that offers a request–response model to a collection of back-end modules. Each module offers a specific computational functionality but is also compatible with neighboring blocks.

The architecture consists of four conceptual layers: a presentation layer, an application control layer, a computational processing layer and an output generation layer. The presentation layer offers file upload and user interface. The application control layer executes the modules. The computational layer offers geometry processing, cost modeling, path optimization, nesting optimization, and AI-based recommendation. The output layer offers PDF quotations and visualizations.

4.1 Overall System Flow

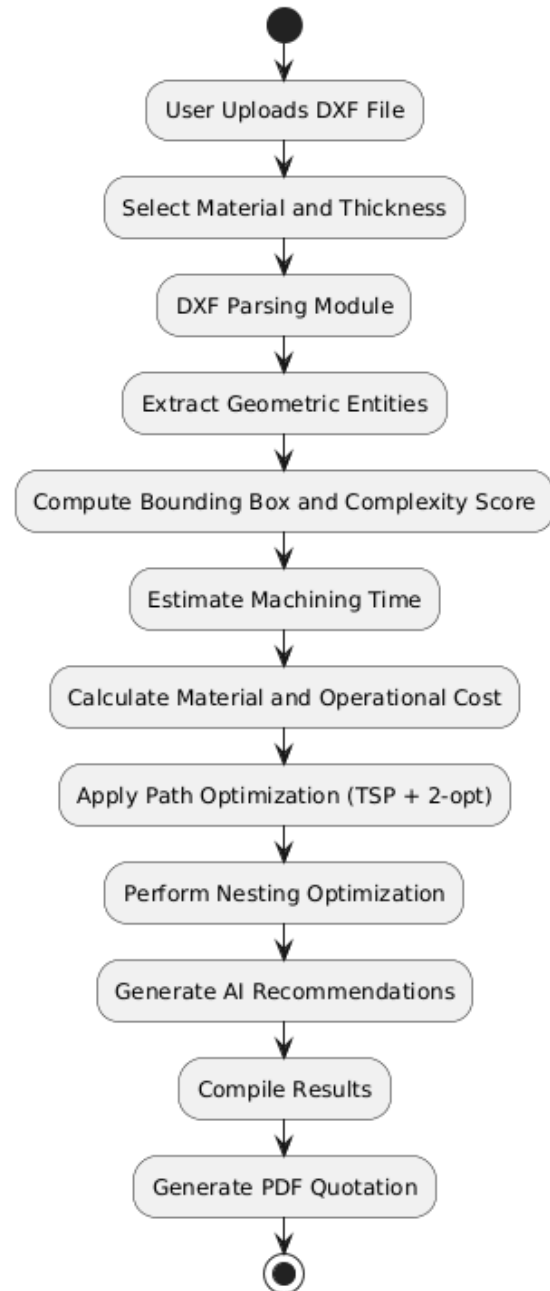


Fig. 1 System Flow

4.2 Layered Architectural Model

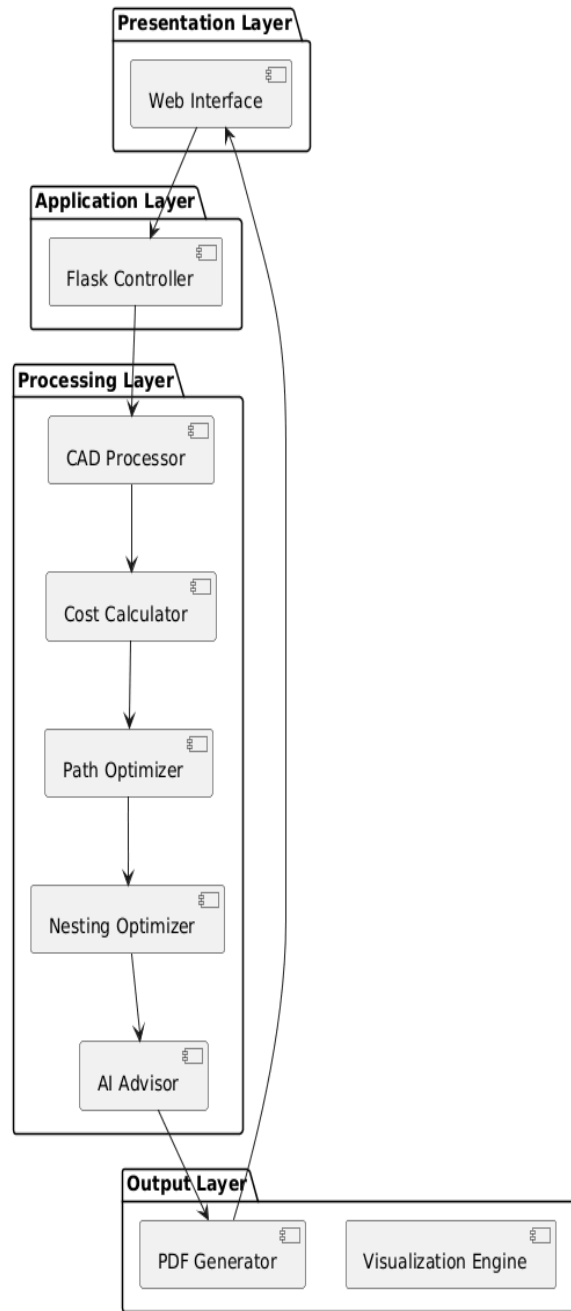


Fig. 2 Architectural Model

4.3 Architectural Description

A web-based part design system has been developed that encompasses cost estimation and manufacturing optimization. The system takes a DXF file and material parameters such as thickness and cost per square unit from a web interface. The system is built with a Flask controller that handles all incoming requests and calls multiple back-end modules. The CAD processor extracts a DXF file, which is then used to calculate lengths and densities of geometric entities such as lines, arcs, splines, and circles in addition to other geometric metrics that are used to compute a geometric complexity score.

These parameters are then passed to the cost calculation module which converts the total cutting length and thickness to machining time based on user-provided feed rates. The total material cost, operational cost, and tax are combined to produce a cost breakdown.

The path optimizer then optimizes the raw cutting path to minimize non-productive tool travel by solving a Traveling Salesman problem using a heuristic algorithm that is refined with a local search algorithm. The nesting optimizer estimates how well the parts could be arranged within sheet boundaries to estimate its material utilization efficiency. The AI advisory module can then produce recommendations with respect to manufacturability and material usage based on the provided geometric parameters and cost metrics. Finally, a PDF generator module creates a structured quotation file for the user that includes cost breakdowns and the improvements made by the various optimization algorithms.

The use of multiple modules allows for the system to be scalable and extensible. Various optimization algorithms can be replaced or improved without affecting the functionality of the entire system.

5. Proposed Methodology

This paper describes a proposed CNC AI Quotation Generator that transforms the raw DXF geometry into an intelligent and polished machining quotation by a sequential chain of computational steps. The proposed method integrates geometric assessment, complexity modeling, cost estimation, combinatorial optimization and advisory intelligence into a unified framework. The mathematical and algorithmic modelling of each step is proposed to guarantee repeatability, automation and scalability.

5.1 Geometric Extraction and Entity Measurement

The initial step is to parse the DXF file and to extract the various geometric entities: lines, arcs, circles, polylines,

splines and ellipses. The geometric length of each entity is derived by analytical or approximate methods.

For a line segment between two points (x_1, y_1) and (x_2, y_2) , the Euclidean length is computed in :

$$L_{line} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad - \quad 5.1.1$$

For a circular arc with radius r and angle difference θ (in radians):

$$L_{arc} = r \cdot \theta \quad - \quad 5.1.2$$

For a full circle:

$$L_{circle} = 2\pi r \quad - \quad 5.1.3$$

Polyline lengths are computed by adding up lengths of adjacent segments. Spline are sampled at discrete points, and ellipses are approximated with Ramanujan's approximation for ellipse perimeters

$$L_{ellipse} \approx \pi[3(a+b) - \sqrt{(3a+b)(a+3b)}] \quad - \quad 5.1.4$$

The total cutting length is defined as:

$$L_{total} = \sum_{i=1}^n L_i \quad - \quad 5.1.5$$

where L_i represents the length of each geometric entity.

5.2 Complexity Scoring Model

Measuring the complexity of manufacturing.

A weighted scoring function of geometric diversity, entity density and curve existence is used to measure the manufacturing complexity. The complexity score was normalized to a 0-100 scale.

$$C = w_1 N_e + w_2 N_c + w_3 D + w_4 V \quad - \quad 5.2.1$$

where:

- N_e represents total entity count
- N_c represents curved entity count
- D represents geometric density within bounding area
- V represents entity type variation
- w_1, w_2, w_3, w_4 are normalized weights

The final score is scaled:

$$C_{final} = \min \left(100, \frac{C}{C_{max}} \times 100 \right) \quad - \quad 5.2.2$$

This score influences machining time estimation and advisory insights.

5.3 Machining Time and Cost Estimation Model

Machining time is estimated based on total cutting length and material-specific feed rate F_m :

$$T_{machining} = \frac{L_{total}}{F_m} \quad - \quad 5.3.1$$

Material volume for sheet cutting is estimated as:

$$V = L_{total} \times t \quad - \quad 5.3.2$$

where t is material thickness.

Material cost is calculated as:

$$Cost_{material} = V \cdot R_m \quad - \quad 5.3.3$$

where R_m is material cost per unit volume.

Machine operational cost is:

$$Cost_{machine} = T_{machining} \cdot R_h \quad - \quad 5.3.4$$

where R_h is machine hourly rate.

Total cost before tax:

$$Cost_{base} = Cost_{material} + Cost_{machine} + Cost_{setup} \quad - \quad 5.3.5$$

Final quotation cost including tax (e.g., GST at 18%):

$$Cost_{final} = Cost_{base} \times (1 + 0.18) \quad - \quad 5.3.6$$

5.4 Cutting Path Optimization

Cutting path sequencing is expressed as a traveling salesman problem in which each entity centroid is a node of a graph, and the goal is to reduce total traversal distance.:

$$\min \sum_{i=1}^{n-1} d(v_i, v_{i+1}) \quad - \quad 5.4.1$$

where $d(v_i, v_j)$ represents Euclidean distance between entity centroids.

The optimization process consists of two stages:

Firstly, an initial feasible route is generated by a Nearest Neighbor heuristics. Secondly, a 2-opt improvement process iteratively exchanges edge pairs to shorten the path until no further improvement can be achieved.

The optimized path length L_{opt} is compared to the original sequence L_{orig} to compute efficiency gain:

$$Efficiency\ Gain = \frac{L_{orig} - L_{opt}}{L_{orig}} \times 100 \quad - \quad 5.4.2$$

5.5 Nesting Optimization

The utilization of material is approximated by a bin packing algorithm on a grid. If A_{part} area of the part's bounding box and A_{sheet} area of the sheet, then utilization is:

$$Utilization = \frac{A_{part}}{A_{sheet}} \times 100 \quad - \quad 5.5.1$$

The Nesting algorithm attempts to reduce the unused area of the sheets, while respecting spacing constraints between parts.

5.6 AI Advisory Mechanism

A Geometric and Cost Metadata based AI Advisory System for Manufacturing Process Optimization. The architectural design of an AI advisory module that ingests geometric and cost metadata to provide contextual recommendations. The input features used are:

- Total cutting length
- Complexity score
- Material type
- Thickness
- Optimization gains

These are structured into prompts for a language model to generate manufacturability, material and cost recommendations. This approach turns a deterministic cost estimation model into a decision support system.

6. Experimental Results

To assess the performance of the proposed CNC AI Quotation Generator, a series of DXF design files with varying geometric complexity were tested with various materials and thicknesses. The experimentation evaluated the following metrics: estimation accuracy, processing time consistency, path optimization gain, and material utilization improvement.

6.1 Experimental Setup

The proposed system was evaluated with a set of two-dimensional DXF files composed of lines, circles, splines, arcs and polylines. Test cases were grouped into three categories: low, medium, and high complexity based on the number of entities and curve density. The materials were selected with typical feed rates and cost per unit. The materials tested were steel, aluminum, and brass.

The test cases were executed on a typical workstation environment in Python with Flask as the backend. The processing time was measured from the file upload to the completion of the quotation.

6.2 Cost Estimation Consistency

To demonstrate the estimation consistency, the results of the system generated cost of machining was compared against manually generated cost with the same feed rate and cost per unit. The cost estimation of the proposed system was found to be consistent with the manual estimation, with only minor difference due to rounding. While the traditional manual estimation process requires re-estimation for each geometry modification, the proposed system is deterministic and provides consistent estimation.

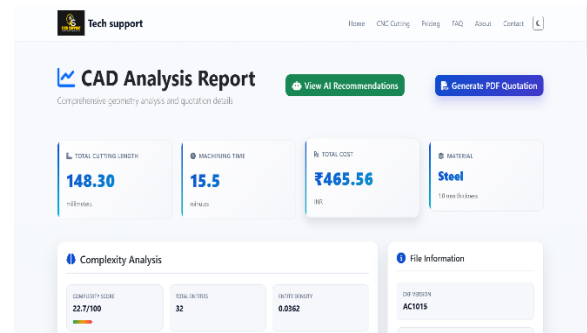


Fig. 3 Cost Estimation Report

6.3 Path Optimization Performance

Tool path optimization performance was quantified by comparing the original traversal length with the optimized path length produced by the Nearest Neighbor and 2-opt refinement algorithm.

The percent improvement was calculated by equation - 5.4.2.

Total nonproductive travel distance decreased with path optimization, across medium and high complexity test cases. In high entity cases, the 2-opt path refinement substantially improved the initial heuristic solution, revealing the connectivity between machining efficiency and path optimization. Less travel is directly proportional to less machining time and increased productivity.

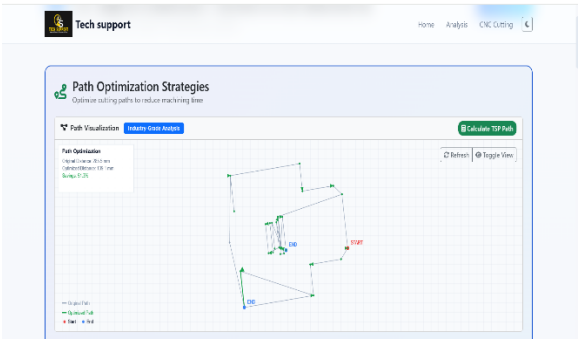


Fig. 4 Path Optimization Graph

6.4 Material Utilization Improvement

The nesting module calculated sheet utilization percentages for multiple part dimensions. Material utilization was computed as in equation – 5.5.1.

Experimental data indicated more efficient material use than what would result from a naïve placement hypothesis. For slowly increasing bounding geometries, efficient layout practices resulted in significant reductions of estimated material waste. This translates directly to cost savings in sheet-based manufacturing processes.

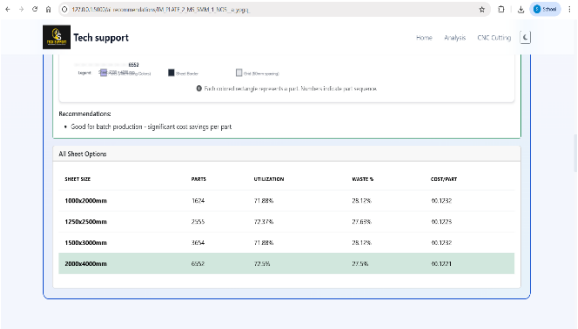


Fig. 5 Material Utilization Report

6.5 Processing Time Analysis

The performance of the developed system is evaluated in terms of total quotation generation time, for different DXF files, even for complex geometry. The geometry extraction and cost estimation parts are marginal in terms of generation time, as they do not involve heavy computation. The overall path optimization time is high, since the heuristic employed for this part is quadratic in the number of geometry entities. However, the time taken to execute the path optimization is not prohibitive for typical industrial design.

The addition of AI advisory module introduces a small amount of latency based on the response time of the API. However, it does not block the quotation generation procedure.

7. Conclusion and Future Work

This paper proposed an algorithmic framework for CNC AI Quotation Generator, that automates the generation of machining quotations. The framework combines various algorithmic steps ranging from geometric analysis, cost estimation based on machining time, cutting path optimization, and material utilization optimization with AI advisory. Our contribution includes a framework that uses a deterministic cost model for machining time and cost estimation, combinatorial path and utilization optimization, and AI based human-in-loop advisory. Experimental results of cost generation for different DXF files confirm that the cost estimates are consistent. Further, the developed path optimization algorithms result in non-productive tool travel and projected material utilization reduction, as compared to the paths generated by the quoter. Studying the overall effectiveness of the proposed work will be taken into consideration in the future work.

This paper has presented a modular architecture and an algorithm for developing an artificial-intelligent quotation-

based support system in Computer Numerical Control (CNC) manufacturing. The architecture is scalable, extensible and adaptable to future integration of more sophisticated optimization algorithms and cost modeling techniques. The developed system has implemented a cost model for estimating machining cutting time and a cost model for estimating cost based on finite element method (FEM) predictions of machining forces. The cost models were integrated in the framework to provide a quotation-based support system. Implementation results are also discussed. Future work on architecture and implementation aspects includes a support for 3D CAD modeling data formats such as STEP files, integration of skill-based learning models trained with historic machining data, integration of predictive models for dynamic parameters in machine tool such as tool wear, energy consumption, and integration of the framework in a cloud-based manufacturing context.

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